NUS
National University of Singapore

ME1103
Principles of Mechanics and Materials

PROBLEM SET 4

Question 1

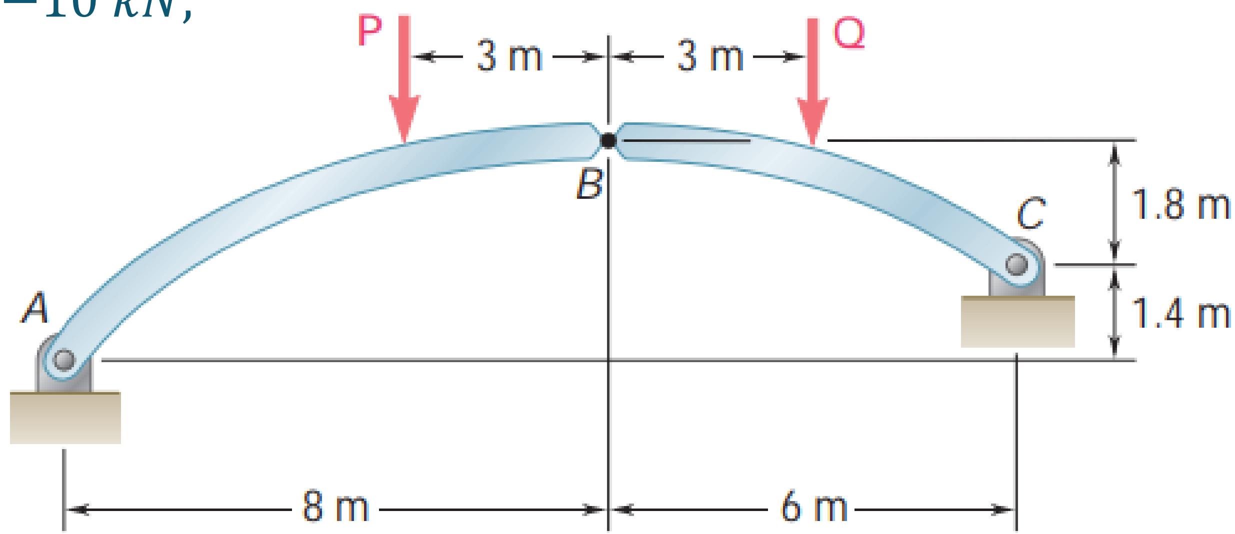
The axis of the three-hinge arch *ABC* is a parabola with the vertex at *B*. Knowing that *P* = 112 *kN* and *Q* = 140 *kN*, determine:

(i) the components of the reaction at *A*, and

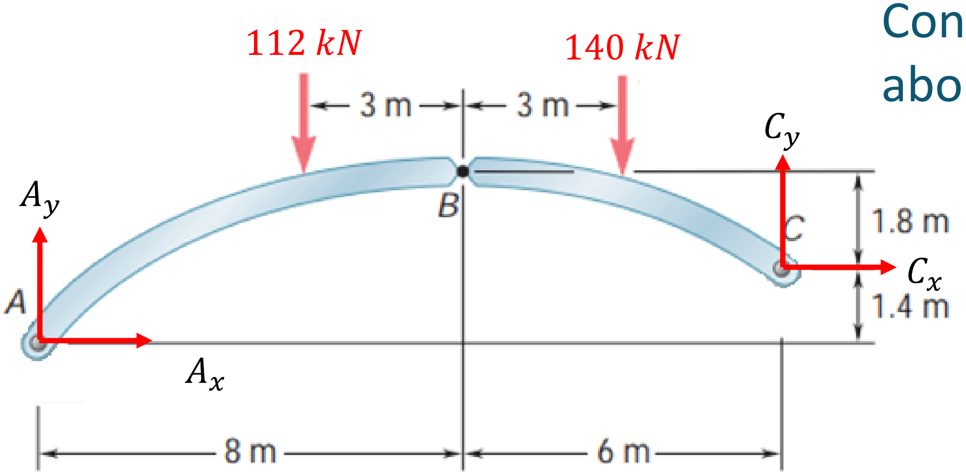
(ii) the components of the force exerted at *B* on segment *AB*.

Answer: $A_x = 200\ kN, A_y = 122\ kN;$

$B_x = -200\ kN, B_y = -10\ kN;$



Question 1…



Consider the entire assembly and taking moment about *C*, gives

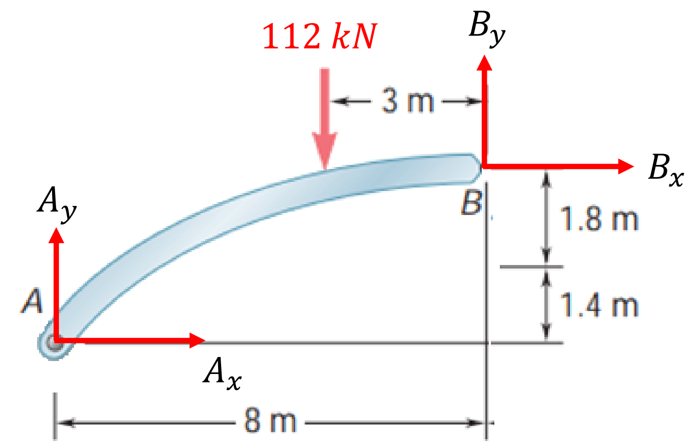
$$A_x(1.4) - A_y(14) + (112)(9) + (140)(3) = 0$$

$$\Rightarrow 14A_y - 1.4A_x = 1428$$

Next, consider member *AB*, and taking moment about *B*, gives

$$A_x(3.2) - A_y(8) + (112)(3) = 0$$

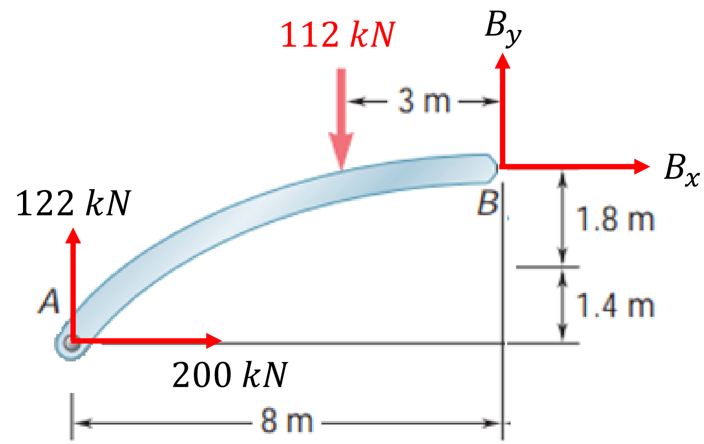
$$\Rightarrow 8A_y - 3.2A_x = 336$$



Solving the two equations, we get

$$A_x = \mathbf{200\ kN}\ and\ A_y = \mathbf{122\ kN}$$

Question 1…



Finally, using the force equations, we get

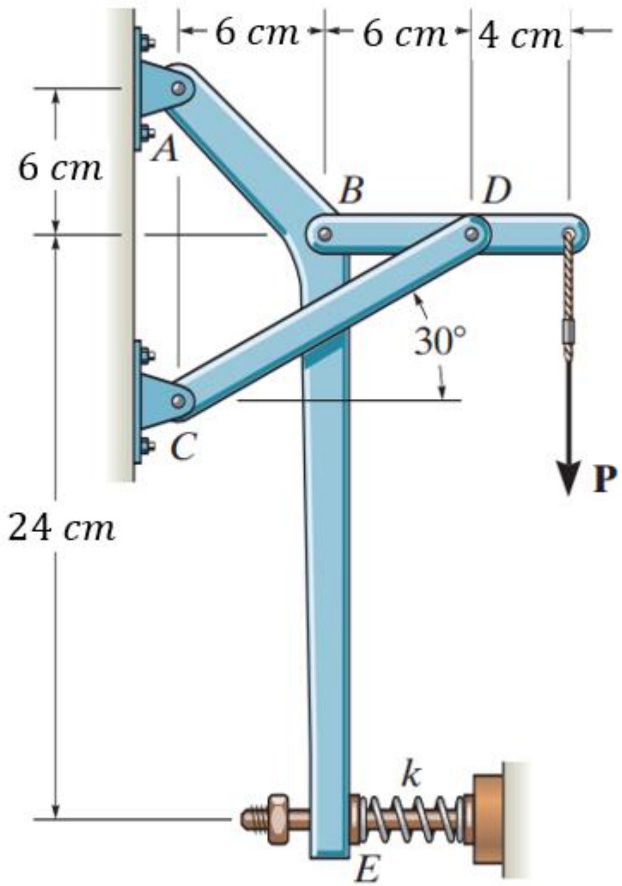
$$\sum F_x = 0, \qquad 200 + B_x = 0 \Rightarrow B_x = \mathbf{-200\ kN}$$

$$\sum F_y = 0, \qquad 122 - 112 + B_y = 0 \Rightarrow B_y = \mathbf{-10\ kN}$$

Question 2

Determine the force P on the cable if the spring is compressed 0.5 *cm* when the mechanism is in the position shown. The spring has a stiffness of k = 10 *kN/m*.

Answer: $P = 70.36\ N$



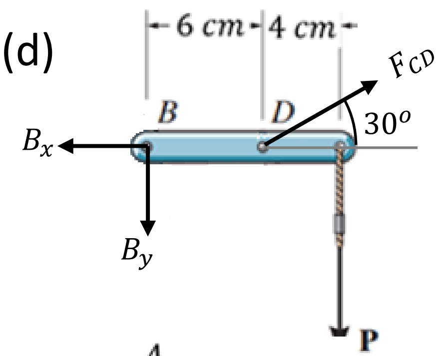
Question 2...

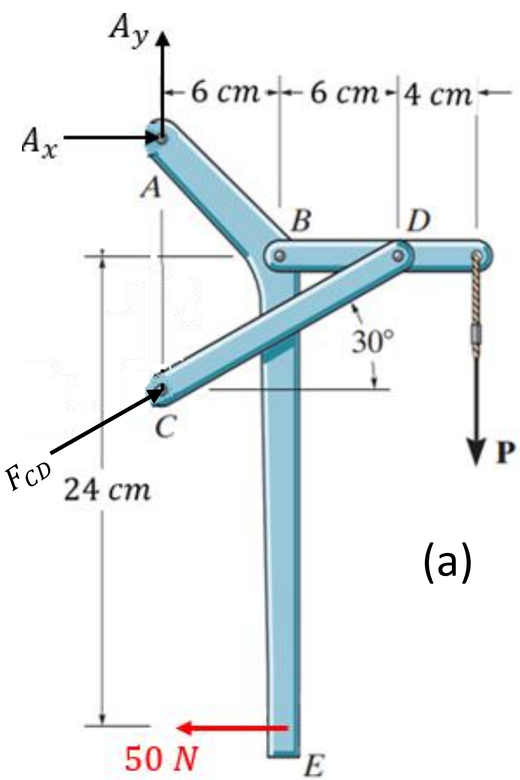
First, the spring force at *E* is

$$F_{spr} = k\Delta l = 10000 \times 0.005 = 50\ \mathrm{N}$$

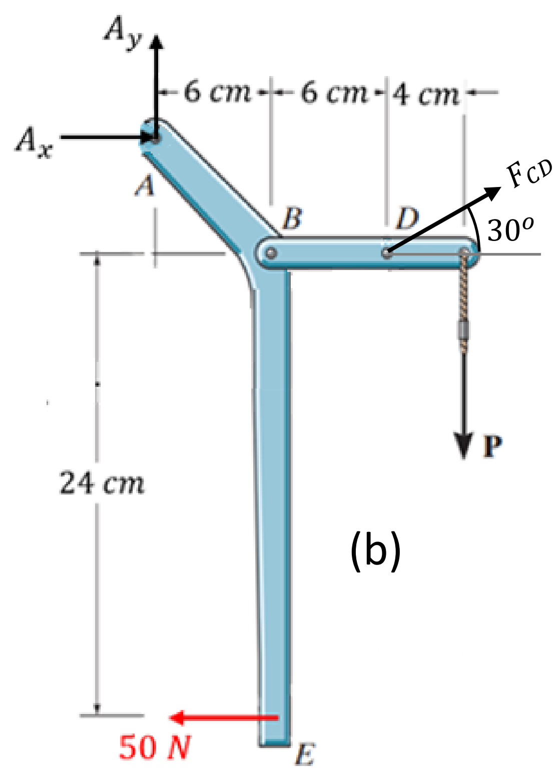
The *FBDs* for this example are summarized as follows.

(d)

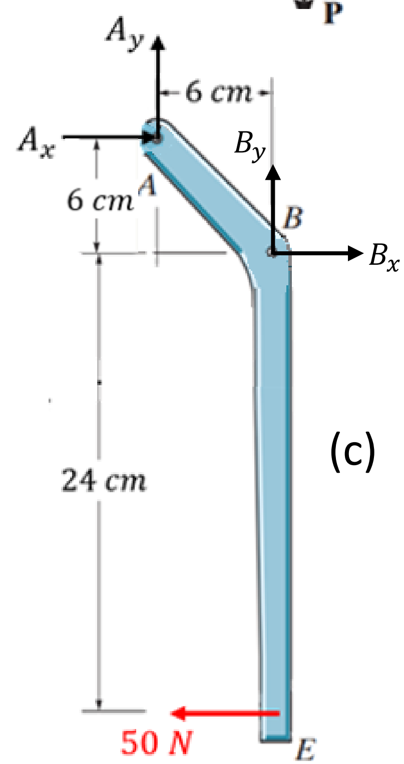




(a)



(b)



(c)

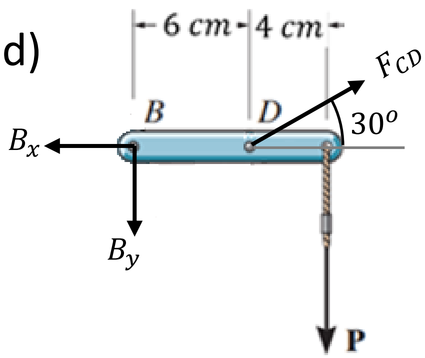
Question 2…

All the *FBDs* have 4 unknowns and hence can be used alone to solve this problem.

FBDs (a) and (b) are "identical" after recognizing member *CD* is a 2-force body.

To solve for *P*, either (b) or (d) must be used. *FBD* (d) is preferred for its simplicity.

(d)



Taking moment about *B*, gives

$$F_{CD} sin30^o(6) - P(10) = 0 \Rightarrow F_{CD} = \frac{10}{3}P$$

And force equilibrium gives

$$-B_x + F_{CD} cos30^o = 0 \Rightarrow B_x = \frac{\sqrt{3}}{2}F_{CD}$$

$$-B_y - P + F_{CD} sin30^o = 0 \Rightarrow B_y = \frac{1}{2}F_{CD} - P$$

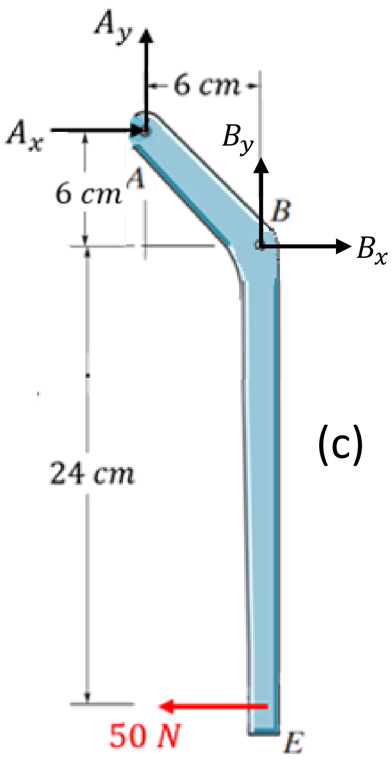
Substituting $F_{CD} = \frac{10}{3}P$ into these expressions, we have

$$B_x = \frac{\sqrt{3}}{2}\left(\frac{10}{3}P\right) = \frac{5}{\sqrt{3}}P, \qquad and \qquad B_y = \frac{1}{2}\left(\frac{10}{3}P\right) - P = \frac{2}{3}P$$

Question 2…

Finally, we need one more equation that only involves $P, F_{CD}, B_x\ and\ B_y$.

Moment equation about A in FBD (b) and (c) generates that outstanding equation. And again for simplicity, FBD (c) is chosen.



(c)

Taking moment about A, gives

$$B_x(6) + B_y(6) - 50(30) = 0$$

$$\Rightarrow B_x + B_y = 250$$

Finally, substituting the expressions for B_x and B_y, gives

$$\frac{5}{\sqrt{3}}P + \frac{2}{3}P = 250$$

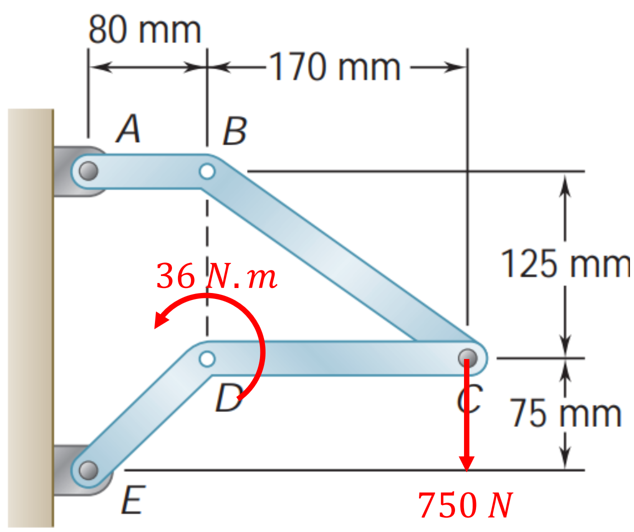
$$\Rightarrow P = 70.355\ N$$

Question 3

Determine the supporting forces at joints *A* and *E* when a 750-*N* force directed vertically downward is applied at joint *C*, and a couple of 36 *N.m* is applied at *D* in counter-clockwise direction.

Find also the inner forces at pin *C* on member *CDE*.

Note: There are two set of inner forces at pin C, one acting on member ABC and the other on CDE. Due to the external load on pin C, these inner forces have different values.



Answer:

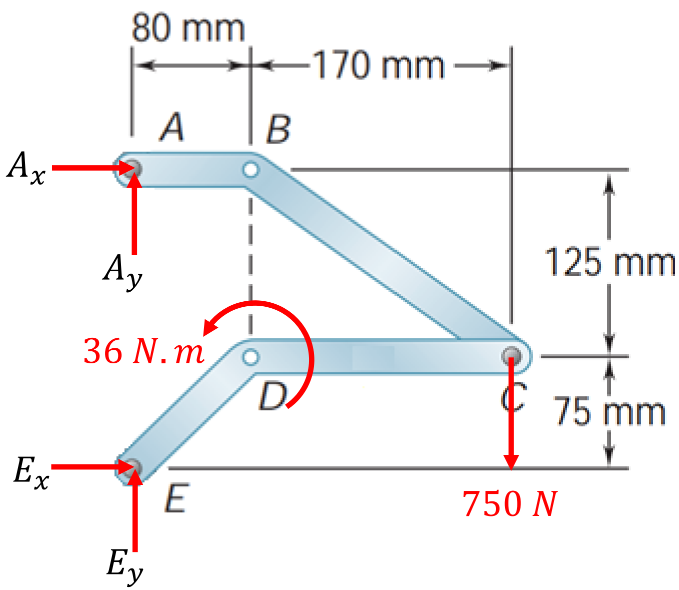
$$A_x = -757.5\ N, A_y = 378.75\ N;$$

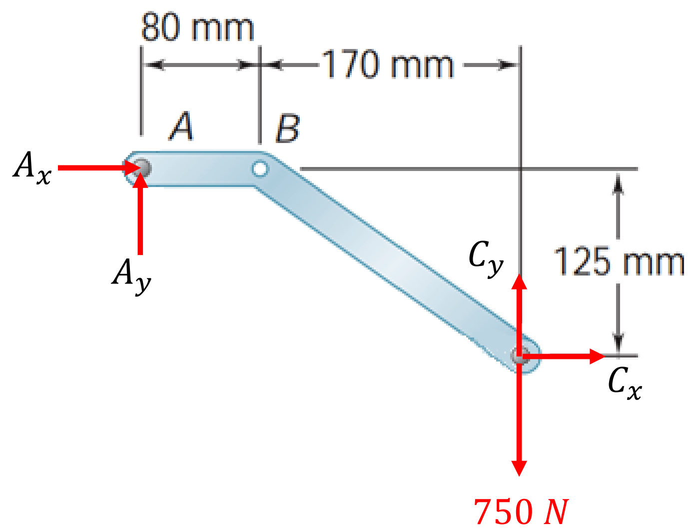
$$C_x = -757.5\ N, C_y = -371.25\ N;$$

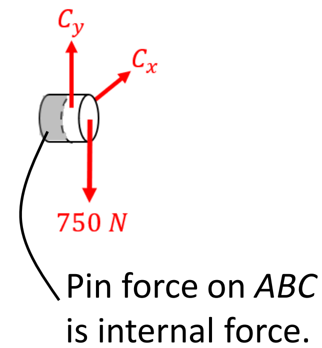
$$E_x = 757.5\ N, E_y = 371.25\ N$$

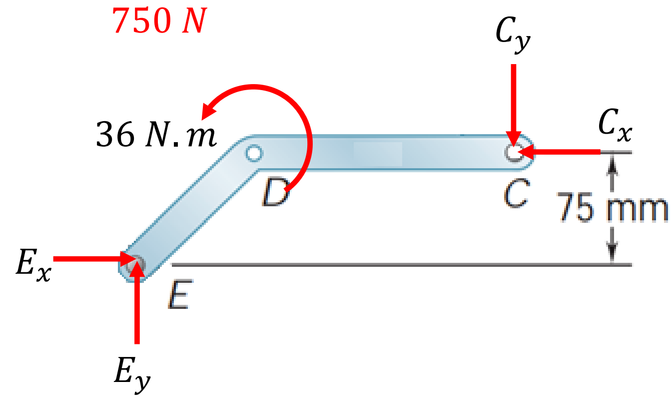
Question 3...

Since the inner force at pin *C* on member *CDE* is required, the pin should go with member *ABC*, so that the inner force can be presented in the *FBD* of its component. With this, the three FBDs are as follows:



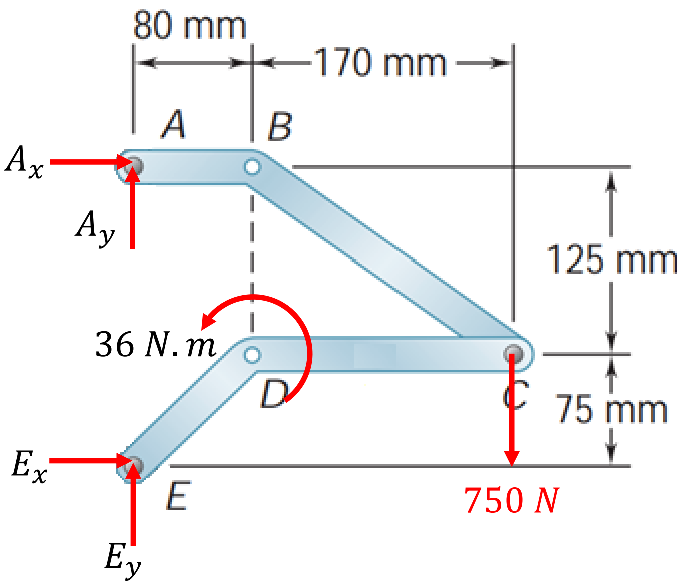






Note: If the pin C goes with member CDE, then the pin force reveals in the FBD would be that acting on member ABC.

Question 3...



Consider the entire system and taking moment about E, gives

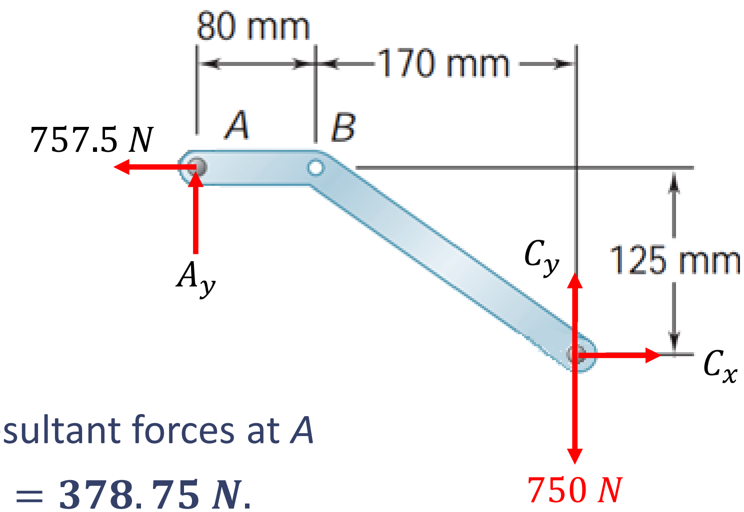
$$-A_x(0.2) - 750(0.25) + 36 = 0$$

$$\Rightarrow A_x = \mathbf{-757.5\ N}$$

Consider member ABC, and taking moment about C,

$$(757.5)(0.125) - A_y(0.25) = 0$$

$$\Rightarrow A_y = \mathbf{378.75\ N}$$



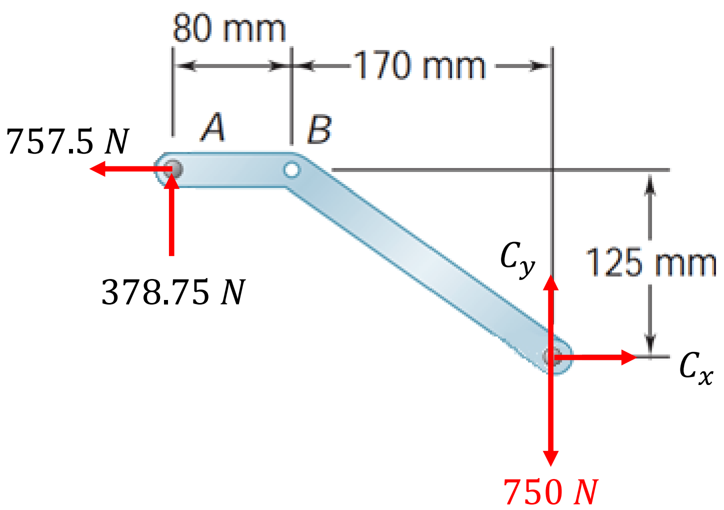
Note: Member ABC is a two-force body. Hence, the direction of the resultant forces at A and C must be along $\vec{r}_{A/C}$. This also means that $\frac{A_y}{A_x} = \frac{125}{250} \Rightarrow A_y = \frac{1}{2}A_x = \mathbf{378.75\ N}$.

Question 3…

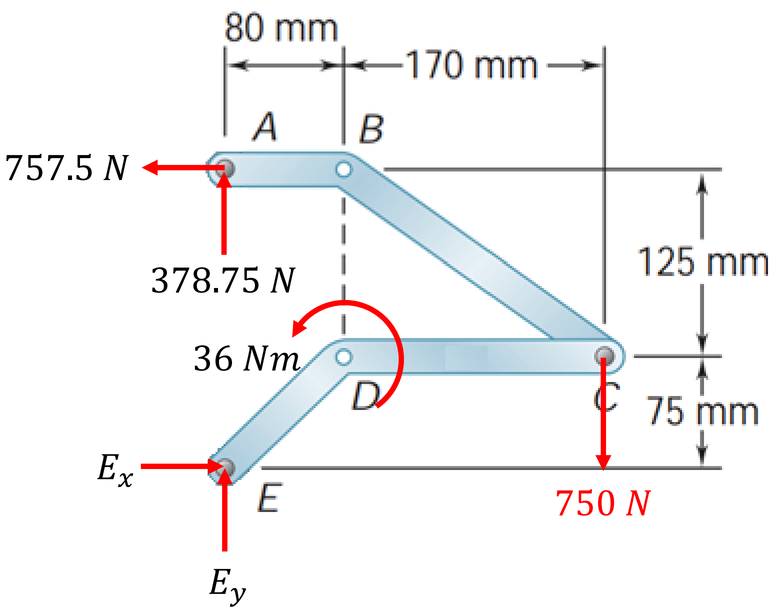
From the force equations, we have

$$-757.5 + C_x = 0 \Rightarrow C_x = \mathbf{757.5\ N}$$

$$378.75\ - 750 + C_y = 0 \Rightarrow C_y = \mathbf{371.25\ N}$$



Now the pin force on *CDE* is the reaction force above, i.e. $C_x = \mathbf{-757.5\ N}$ and $C_y = \mathbf{-371.25\ N}$



Finally, from force equation of the entire system, we get

$$E_x - 757.5 = 0 \Rightarrow E_x = \mathbf{757.5\ N}$$

$$E_y + 378.75 - 750 = 0 \Rightarrow E_y = \mathbf{371.25\ N}$$

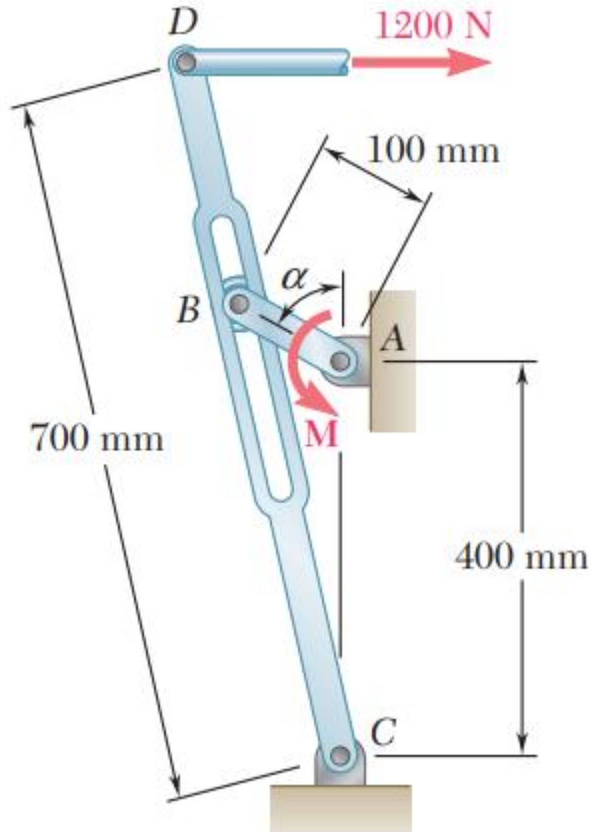
Question 4

The Whitworth mechanism shown is used to produce a quick-return motion of point *D*. The collar at *B* is pinned to the crank *AB* and is free to slide in a slot cut in member *CD*.

Determine the couple *M* that must be applied to the crank *AB* to hold the mechanism in equilibrium when:

(a) $\alpha = 0$, and

(b) $\alpha = 60^o$. At this position, calculate also the reaction forces at *A.*



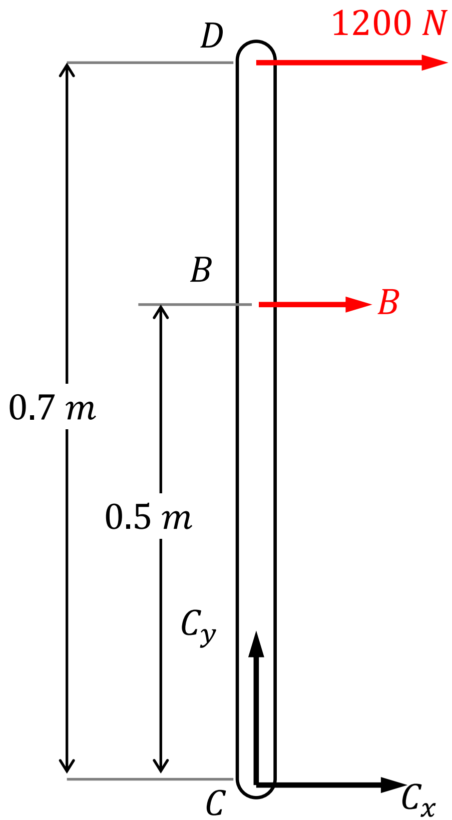
Answer:

(a) $M = 168\ N.m$

(b) $M = 117.8\ N.m,\ \ A_x = -1768\ N, A_y = -340\ N;$

Question 4...

(a) When $\alpha = 0$, i.e. rod *CBD* is vertical, the *FBD* is given by



Note that the reaction at *B* has only horizontal component because it can slide freely vertically.

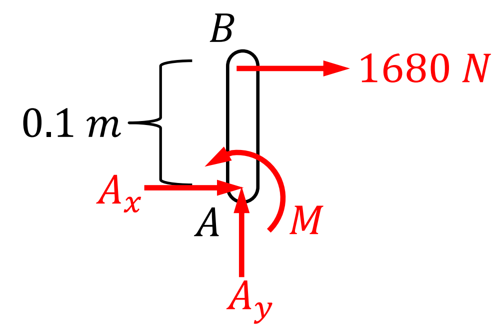
Now, taking moment about *C*, we have

$$-1200(0.7) - B(0.5) = 0 \Rightarrow B = -1680\ N$$

Next, consider the link *AB*, and taking moment about *A*, gives

$$M - 1680(0.1) = 0$$

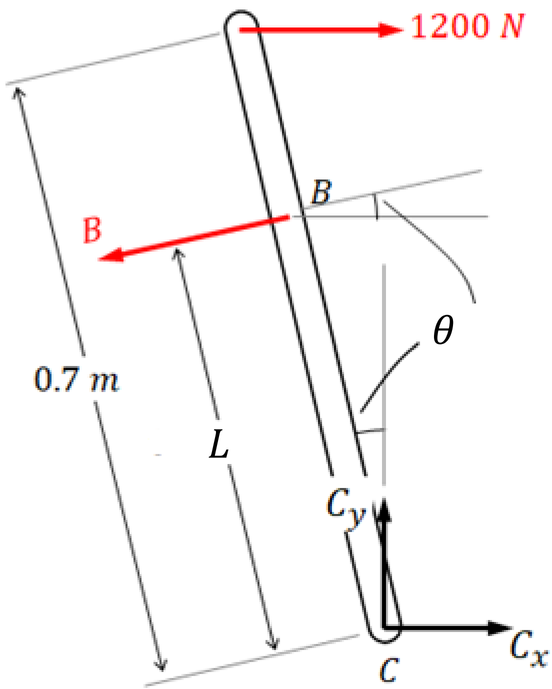
$$\Rightarrow M = \mathbf{168\ N.m}$$

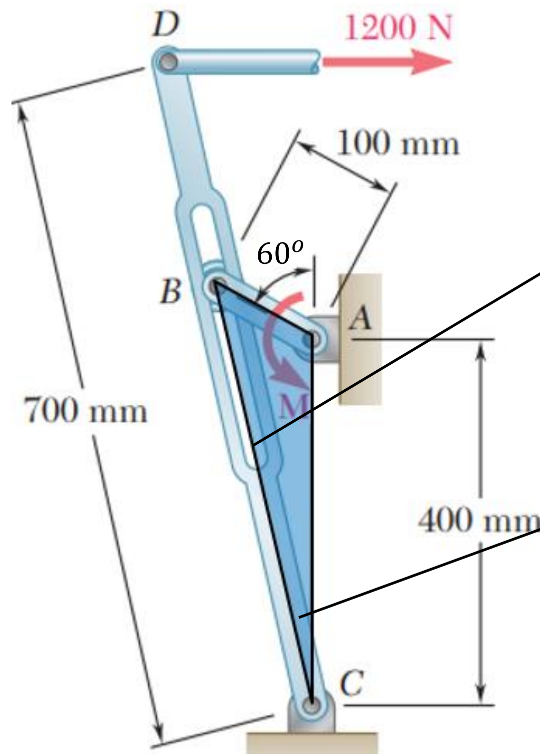


Note: AB is not a two-force body because of the applied couple M.

Question 4…

Next, consider the *FBD* of *CBD*.





$$BC^2 = (0.4)^2 + (0.1)^2 - 2(0.1)(0.4)\cos 120^o$$

$$\Rightarrow BC = 0.4583\ m$$

$$\frac{\sin\theta}{0.1} = \frac{\sin 120^o}{0.4583}$$

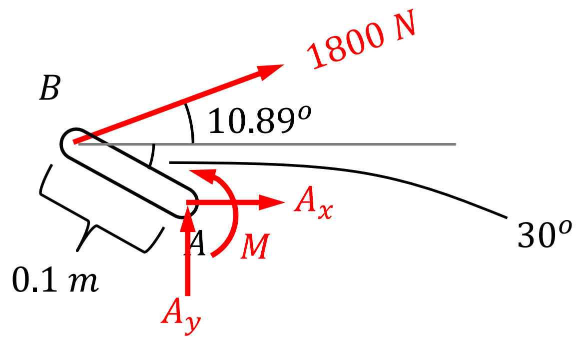
$$\Rightarrow \theta = 10.89^o$$

Taking moment about *C*, gives

$$-1200(0.7\cos 10.89^o) + B(0.4583) = 0 \Rightarrow B = 1800\ N$$

Question 4…

Finally, consider the *FBD* of link *AB.*



From the moment equation about *A*, gives

$$M - 1800(0.1 sin 40.89^o) = 0 \Rightarrow M = \mathbf{117.8\ N.m}$$

Finally, force equilibrium gives

$$A_x + 1800 cos 10.89^o = 0 \Rightarrow A_x = \mathbf{-1768\ N}$$

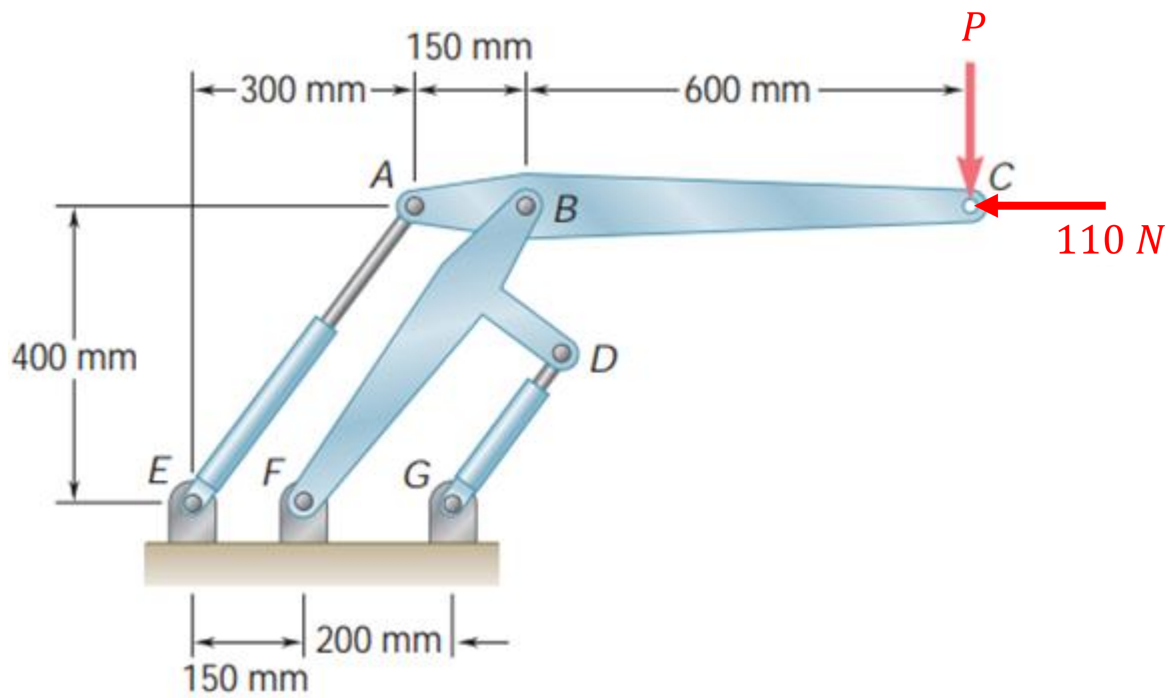
$$A_y + 1800 sin 10.89^o = 0 \Rightarrow A_y = \mathbf{-340\ N}$$

Question 5

Two hydraulic pistons control the position of the link *ABC*. In the static equilibrium position shown, the piston are parallel. Suppose piston *AE* is pulling with $600\ N$, determine the vertical force *P* applied at *C* to arm *ABC,* and the corresponding piston force F_{DG} that hold the assembly at this position.

Answer:

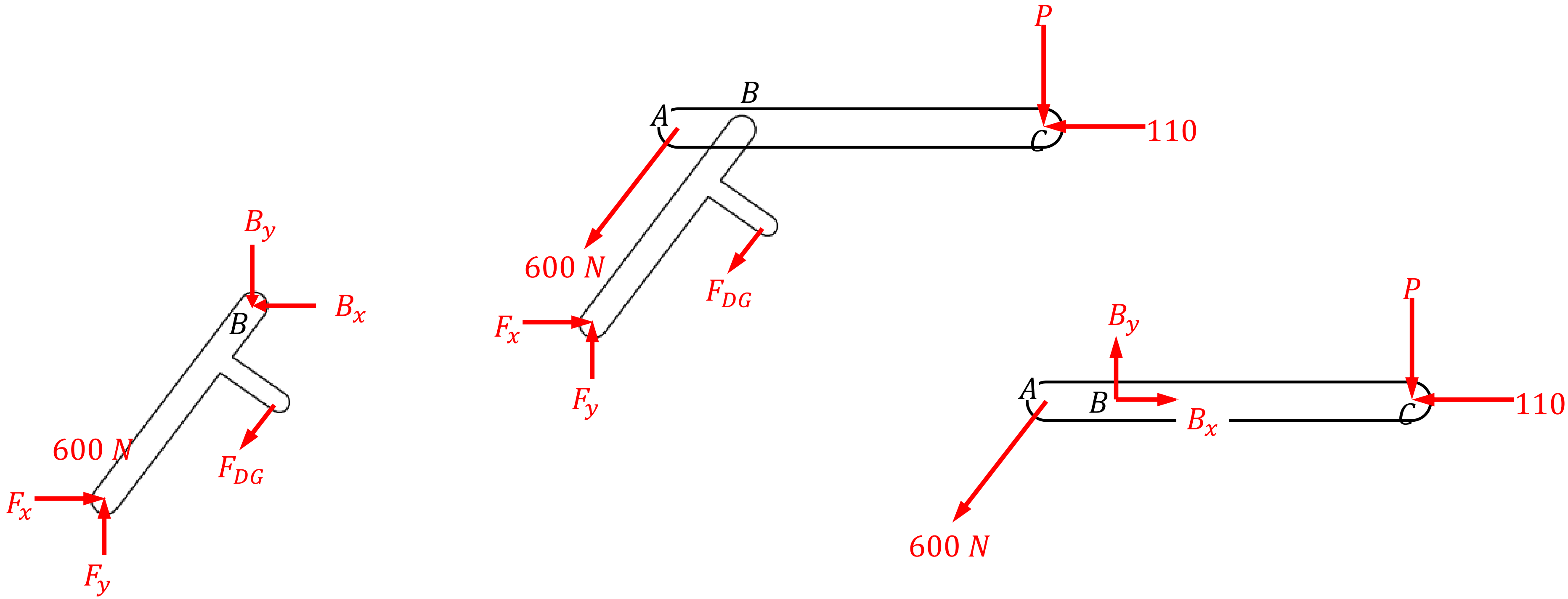
$$P = 120\ N;\ F_{DG} = 50\ N$$



Question 5…

First, both the pistons are two force members, and hence the directions are along $\overrightarrow{AE}$ and $\overrightarrow{DG}$, respectively.

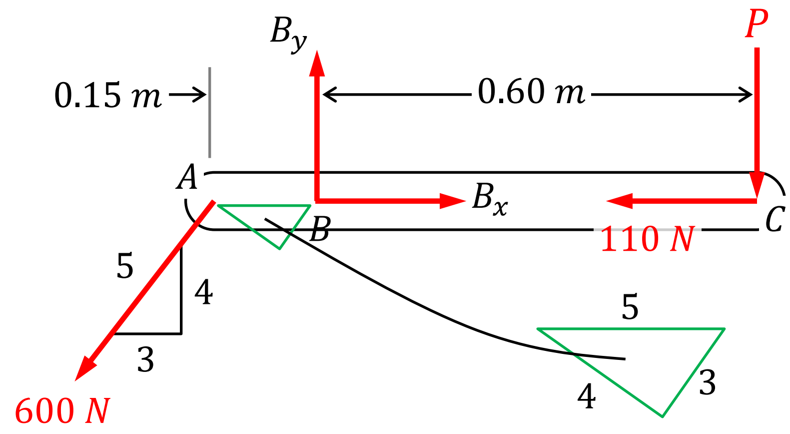
The following three *FBDs* can be identified.



Can you find two equations that only involves the two unknown forces?

Question 5...

From the arm *ABC, we can solve for P*.



Taking moment about *B*, we have

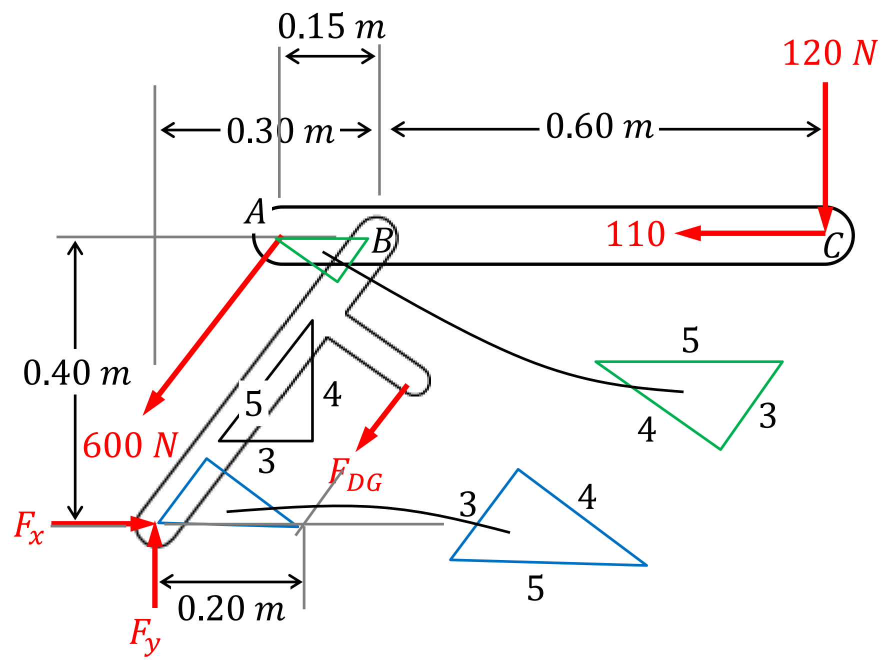
$$\frac{4}{5}(600)(0.15) - P(0.6) = 0 \Rightarrow P = 120\ N$$

Next, from the entire assembly, and taking moment about F, gives

$$(600)\frac{4}{5}(0.15) - F_{DG}\frac{4}{5}(0.2) + 110(0.4)$$

$$-(120)(0.9) = 0$$

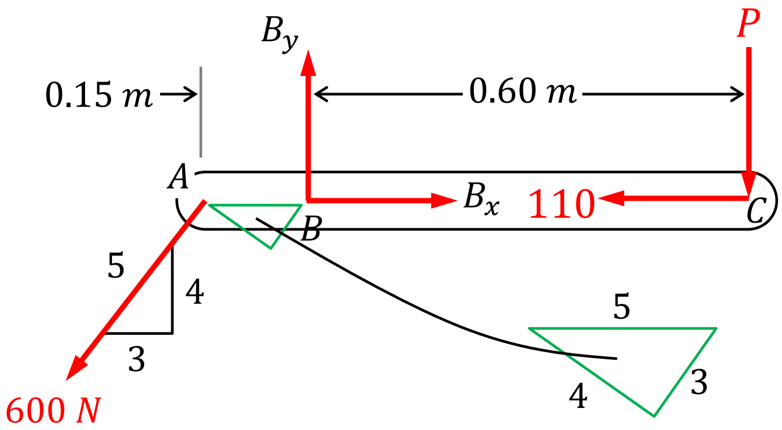
$$\Rightarrow F_{DG} = 50\ N$$



Question 5...

Alternatively, we can solve the problem by looking at the two components separately.

From the arm *ABC*, we get



Taking moment about *B*, we have

$$\frac{4}{5}(600)(0.15) - P(0.6) = 0 \Rightarrow P = 120\ N$$

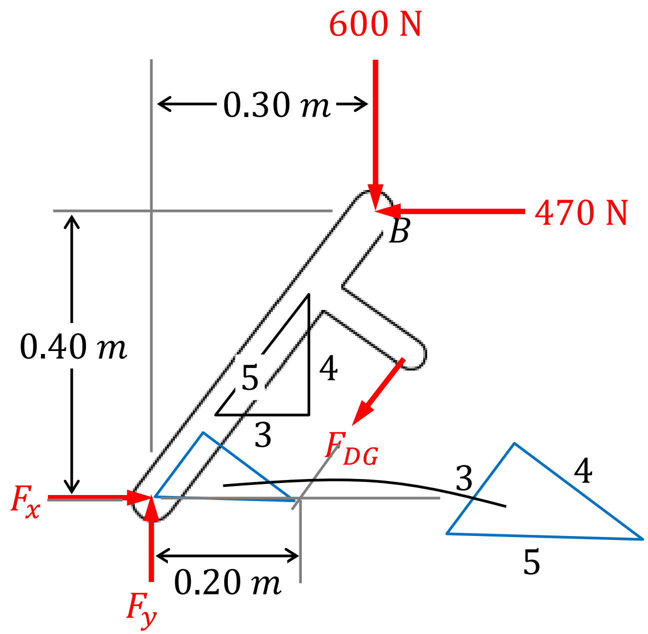
And the force equations are

$$-\frac{3}{5}(600) + B_x - 110 = 0 \Rightarrow B_x = 470\ N$$

$$-\frac{4}{5}(600) + B_y - 120 = 0 \Rightarrow B_y = 600\ N$$

Question 5...

Next, looking at member *BDF*.



Taking moment about *F*, gives

$$-F_{DG}\frac{4}{5}(0.2) + 470(0.4) - (600)(0.3) = 0 \Rightarrow F_{DG} = 50\ N$$